

that isn't easily available and impossibly expensive. Back in 2010, we'd done a test with a recorder with a flat response microphone for testing speakers. A software utility which records frequency responses and plots graphs was used to analyse a frequency sweep sample beginning at a rumbling 20 Hz and going right up to a screeching 15 kHz. Unfortunately, the equipment and environment wasn't sensitive enough to provide too much variance between different speakers.

Which is why for the sake of comparison, you need an industry standard "reference". A speaker or headphone that is known/proven to be flat response. Flat response is a simple concept to understand. It means that the frequency curve produced by the speaker or headphone is flat (or near flat). That means no frequency range – lows (bass) or highs (treble) – is favoured over the other.

But what if you prefer one frequency range – be it bass or treble – over the other? If the speaker is capable of reproducing audio true-to-source, all you need to do is alter the sound going to the speaker. Or you can use digital sound processing (or DSP) to enhance bass! Want to subdue mids to get crunchier guitars? Go for a U-shaped EQ setting. Whatever you do, the speaker will reproduce it faithfully. This is the reason purists are frequently at loggerheads with "audiophiles" who prefer their speakers to be permanently tuned to output a particular type of sound (or "signature" as they call it).

By now, you must have observed that despite our best effort to avoid doing so, we've ended up using quite a lot of jargon. These terms are unavoidable when you step into the world of audio, hence the next section on interpreting reviews will shed much needed light on the subject and de-mystify many things for you.

Interpreting reviews

Because empirical testing is difficult (believe us, we've tried), audio reviews tend to be full of ambiguous adjectives. And in instances when there's no ambiguity, there will most certainly be plenty of jargon.

Case in point:

"The latest high-impedance, low-sensitivity headphones under review were difficult to drive and necessitated an amp, but the sound signature was devoid of

any colouration; neither bright nor warm, just neutral. The soundstaging is phenomenal and the entire frequency range is reproduced accurately – from deep lows to extreme highs – the drivers were pristinely accurate".

Say what?

Understanding terminology

Impedance: "Impedance" is the resistance against electrical flow. In the world of audio gear too, it means the same thing. Measured in Ohms, it basically denotes how difficult it is to move the speaker's driver. By and large, high impedance systems require more power to drive them, hence requiring an amplifier to enhance the signal.

Sensitivity: Another often seen term in the specifications of audio systems (speakers or headphones) is "sensitivity". Sensitivity is the sound pressure level measurement (loudness) taken at a distance of one meter by applying one watt of power. The sensitivity is measured in decibels (dB). Therefore, all other things being constant a high sensitivity audio system will require less power to sound as loud.

Frequency response: The entire audio signal coming out of a speaker belongs to different frequencies of the audio spectrum. The audio range of speaker systems (and the human ear) lies in the range of 20 Hz to 20,000 Hz. The frequency response of a speaker system denotes the range of the frequencies that it can reproduce effectively.

Highs: This term denotes the reproduction of frequencies belonging to the higher end of the audio spectrum. Generally, anything above 5,000 Hz is classified as high frequency. Certain instruments and vocals pitches – such as violins, female vocals, cymbals, etc – belong to this higher side. If a headphone has good "highs" it will be able to reproduce these sounds well.

Lows: Bass notes or "boom" like sounds belong to lower frequencies in the audio spectrum. These are frequencies measuring 20 Hz to 350 Hz. Kick drums, bass guitar and baritone vocals are low frequency sounds or "lows".

Mids: All sounds in the range of 350 Hz to 5,000 Hz. This range brings the overall feel of the audio track. What

people commonly refer to as treble starts from mids and extends into the highs.

Neutral: A "neutral" sound signature is one that reproduces all frequency ranges appropriately without emphasis on any particular range such as bass, mids or highs. Such audio systems are considered true-to-source.

Warm: A "warm" sounding speaker or headphone is one that's a little partial towards the mid frequencies. The sound produced as a result is a little mellow and soft. Whether or not you like this kind of sound is a matter of preference.

Bright: A "bright" sound signature is one that's produced by audio systems with a proclivity for high frequencies. The sound produced as a result is vibrant and hence called bright.

Colouration: Any component in the chain of audio equipment from the source itself to the speakers that adds emphasis on a particular frequency range is said to "colour" the sound. Colour, as it's used here, can even be given a derogatory connotation similar to "taint". The process is called "colouration". Colouration may sound pleasing based on your preferences, but it's a deviation from accuracy.

Soundstaging: This is a pure listening term and the efficacy of it can be debated. Basically, it denotes the perception of depth and positioning of particular sources of sounds such as instruments and vocalists. Some headphones and speaker systems are thought to reproduce or create the illusion of distinct positions of sources within the spacial environment.

Amp: Short for amplifier, "amp" refers to a device that amplifies or enhances the audio signal level. Amplifiers increase power or voltage or both. Headphone amplifiers are quite popular in India these days. Such amps typically give a power output in mWatts (sometime upto 2W). Amps also add colour to the sound.

Dynamic range: It can be defined as the ratio of the loudest and softest sound that an audio system can reproduce without distortion. Again, the ratio is measured in decibels.

Testing headphones

The first step towards making good buying decisions when it comes to headphones is knowing what type of headphone is best for you. And what's best for you depends solely on your use case